

Appendix to “ C^3 -Partitioning: Beyond Last-Level Cache Capacity Isolation in Arm Mixed-Criticality Systems”

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I. ADDITIONAL ANALYSIS OF THE RK3568 SAME-CORE LATENCY

The main paper uses same-core measurements as a control for the cross-core snoop filter (SF) interference path. On RK3568, the same-core stimulus-count sweep exhibits several latency increases as the number of stimulus addresses grows. We repeat the experiment with performance monitoring unit (PMU) measurements to examine whether address translation may contribute to this behavior.

A. Experimental Setup

The kernel module allocates and continuously retains ordinary 4 KiB pages and selects candidate lines with identical PA[22:0] values. The baseline uses the same number of lines from other module-owned pages that do not satisfy this constraint. The probe core primes and reloads the probe. The same-core condition executes the stimulus accesses on the probe core, while the cross-core condition executes them on another core. The timed final reload and the PMU measurement cover the same access.

We record the reload latency together with L1D_CACHE_REFILL, L1D_TLB_REFILL, and LL_CACHE_MISS_RD, which count level-1 data cache refills, level-1 data translation lookaside buffer (TLB) refills, and last-level cache (LLC) read misses, respectively. Event incidence denotes the percentage of timed reload windows containing at least one occurrence of the corresponding event. The experiment uses five newly allocated and independently retained page pools. Lines report the median of the five allocation-level values, and envelopes span their minimum and maximum. X denotes the number of stimulus addresses and excludes the probe address.

B. PMU Observations

Fig. 1 shows that the same-core candidate begins to record LLC read misses at $X = 4$. The miss incidence and reload latency rise markedly around $X = 6$. Neither same-core condition records an L1D TLB refill for $X < 12$.

From $X = 12$, both same-core conditions record an L1D TLB refill in every reload window. This behavior is consistent with address-translation-path activity contributing to the reload latency at larger stimulus counts. Cache and translation events overlap in this region, so the measurements neither establish causality nor isolate the translation contribution to each latency increase.

TABLE I
RK3568 PROBE-RELOAD METRICS AT $X = 16$. ENTRIES ARE MEDIAN ACROSS FIVE NEWLY ALLOCATED PAGE POOLS. THE L1D, TLB, AND LLC COLUMNS REPORT THE INCIDENCES OF L1D CACHE REFILLS, L1D TLB REFILLS, AND LLC READ MISSES, RESPECTIVELY.

Condition	Cycles	L1D (%)	TLB (%)	LLC (%)
Same candidate	355	100.0	100.0	99.8
Same baseline	87	3.3	100.0	1.5
Cross candidate	291	98.6	0.0	96.9
Cross baseline	44	42.3	0.0	0.1

Table I reports the measurements at $X = 16$. Both same-core conditions record 100% TLB refill incidence, confirming that the recorded latency step coincides with probe-side translation-refill activity at this stimulus count. The cross-core candidate reaches 291 cycles with 96.9% LLC read miss incidence and no probe-side TLB refill.

II. CONTROLLER RESPONSE-TIME VARIATION AT 100 AND 10 Hz

Figs. 2 and 3 extend the 1 kHz evaluation in the main paper to 100 and 10 Hz using the same platforms, interference configurations, and GPOS working-set sweep. Each point is obtained from one 600-s run, so the results are used to examine whether the main timing trends persist at lower release rates.

Both lower release rates preserve the separation observed at 1 kHz. Under inter-core interference, SP and C^3 -Partitioning remain substantially below Baseline and WP as the GPOS working set grows. Under combined interference, C^3 -Partitioning remains below SP on both platforms, with a particularly clear separation on RK3568. The larger absolute response-time variation on RK3568 also remains visible at both rates.

III. LOW-RATE ABLATION AND ALLOCATION SENSITIVITY

The main paper evaluates the two coloring levels and task-color allocation ratios at 1 kHz. Figs. 4 and 5 extend this evaluation to 100 and 10 Hz on RK3588 under combined interference. The configurations, fixed 10 MiB RTOS-local working set, and GPOS working-set sweep remain unchanged. Each working-set point contains one 600-s run.

At both lower release rates, C^3 -Partitioning remains below Baseline and the single-level coloring configurations after the GPOS working set enters the high-pressure region. The main

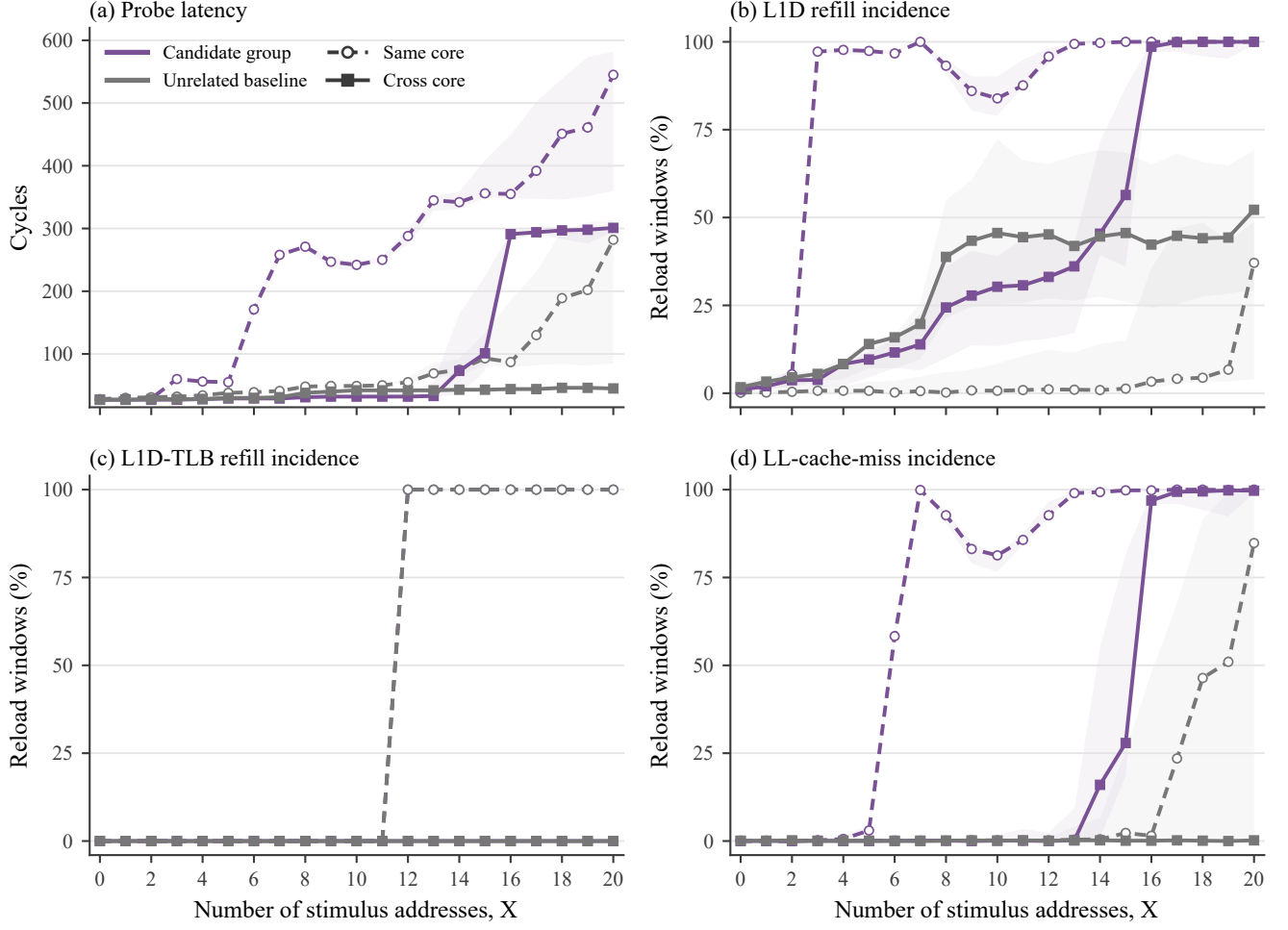


Fig. 1. RK3568 probe-reload latency and PMU measurements across five newly allocated module-owned page pools. (a) Median allocation-level mean latency. (b)–(d) Median allocation-level incidences of L1D cache refills, L1D TLB refills, and LLC read misses in the timed reload window. Both same-core conditions reach 100% TLB refill incidence from $X = 12$. Envelopes span the minimum and maximum across the five allocations. X denotes the number of stimulus addresses and excludes the probe address.

difference from the 1 kHz result appears in the task-color allocation. Assigning 2/4 or 3/4 of the task colors to the critical region lowers $T_{\text{resp}}^{\text{max}}$ relative to the 1/4 allocation and generally reduces controller-window refills and LLC read misses. The 2/4 and 3/4 timing curves remain close, indicating that most of the observed improvement is obtained when the critical allocation increases from 1/4 to 2/4. These results show that the release interval should be considered when selecting the task-color allocation ratio.

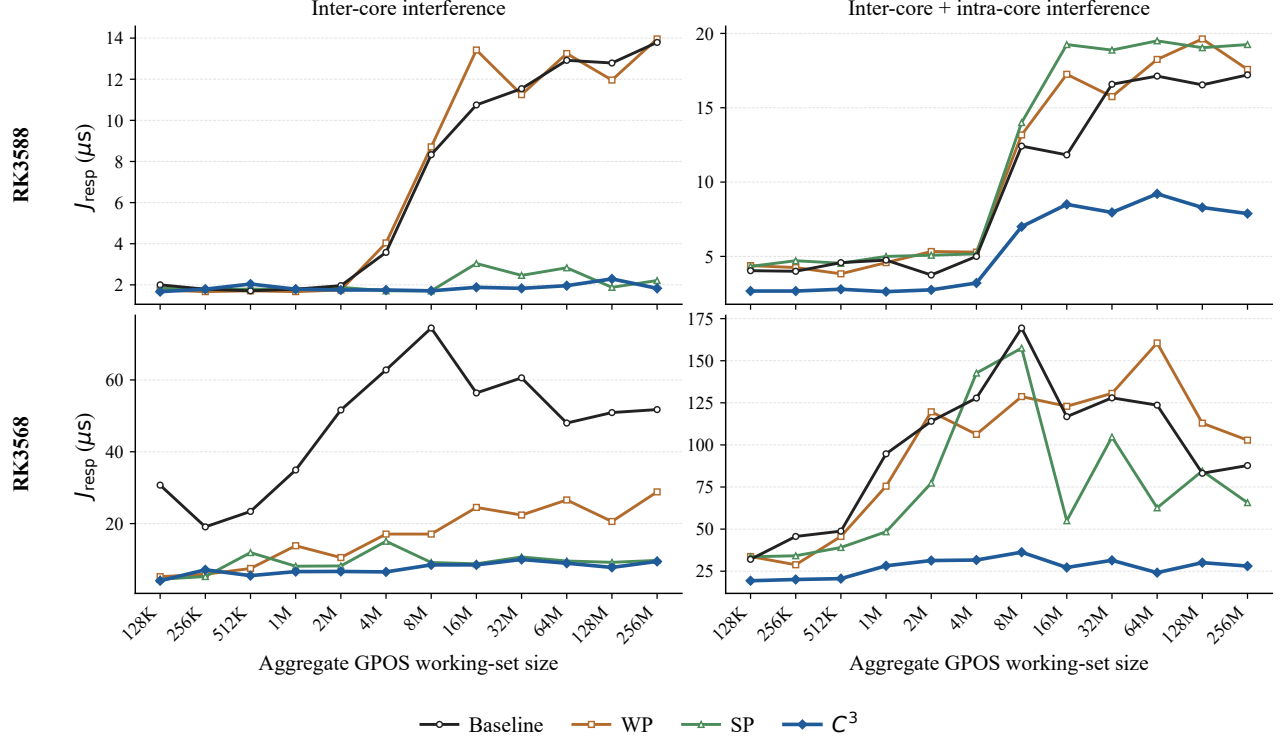


Fig. 2. Peak-to-peak controller response-time variation at 100 Hz across the aggregate GPOS working-set sweep. Combined interference additionally introduces the low-priority memory-intensive task on the RTOS core. Each point is obtained from one complete 600-s run containing 60,000 controller releases.

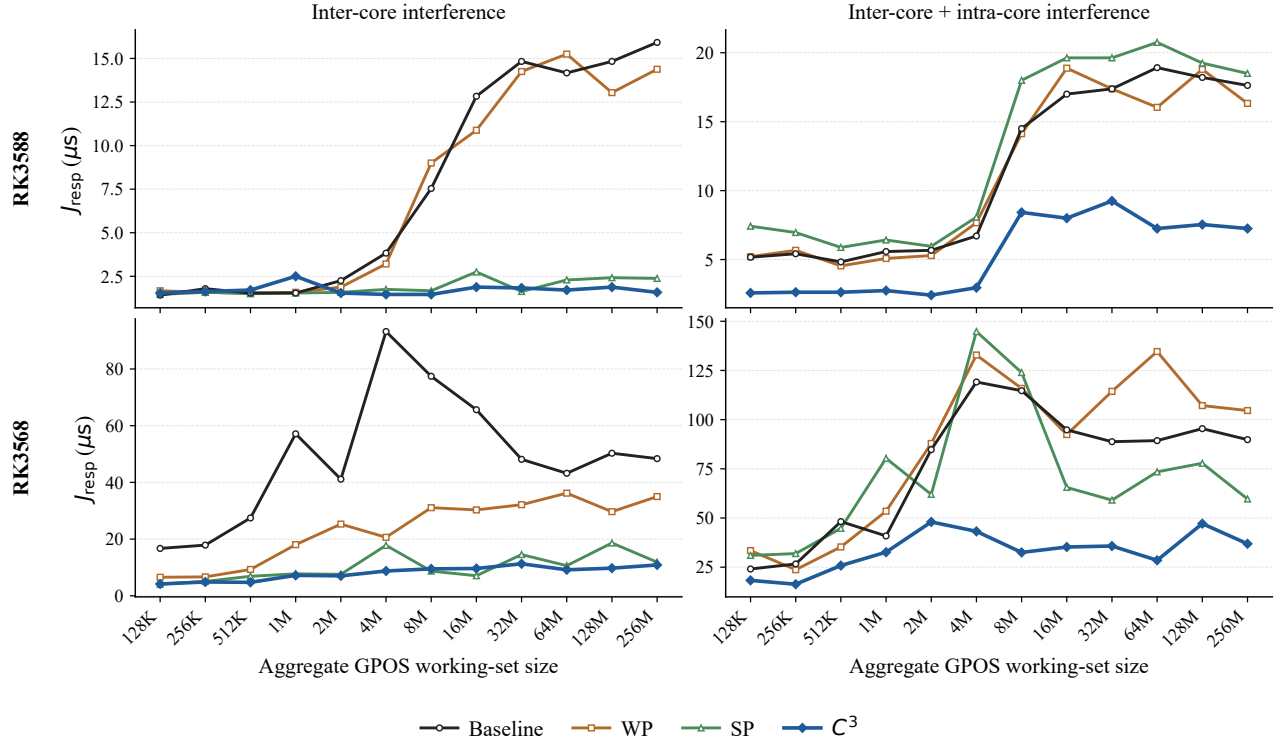


Fig. 3. Peak-to-peak controller response-time variation at 10 Hz across the aggregate GPOS working-set sweep. Combined interference additionally introduces the low-priority memory-intensive task on the RTOS core. Each point is obtained from one complete 600-s run containing 6,000 controller releases.

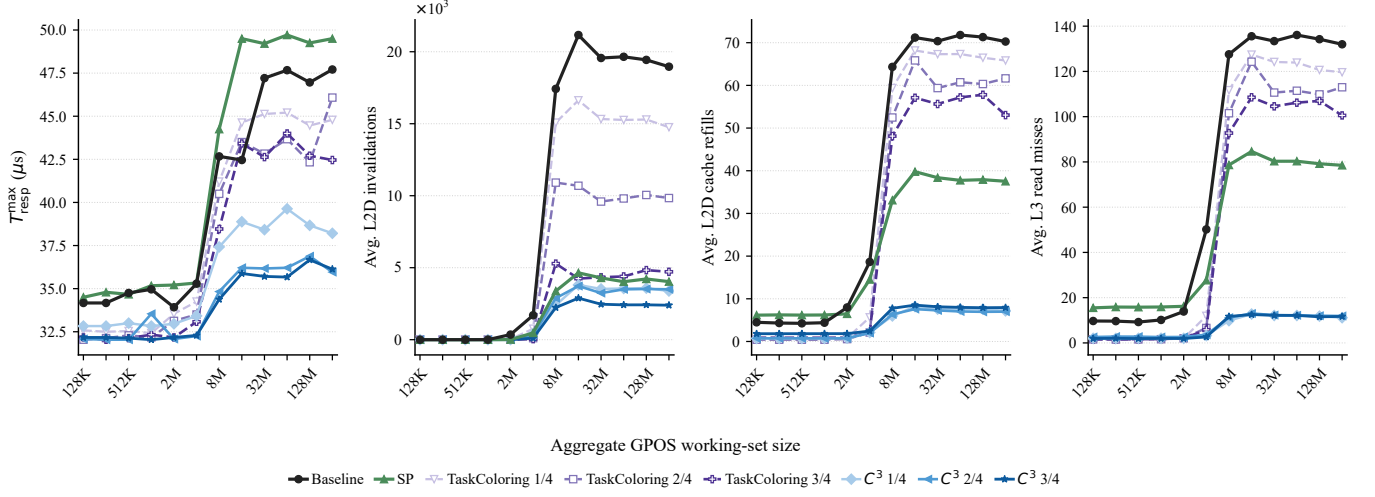


Fig. 4. RK3588 ablation and task-level allocation sensitivity at 100 Hz under combined interference. Fractions denote the proportion of B_T colors assigned to the critical region. Each point is obtained from one complete 600-s run.

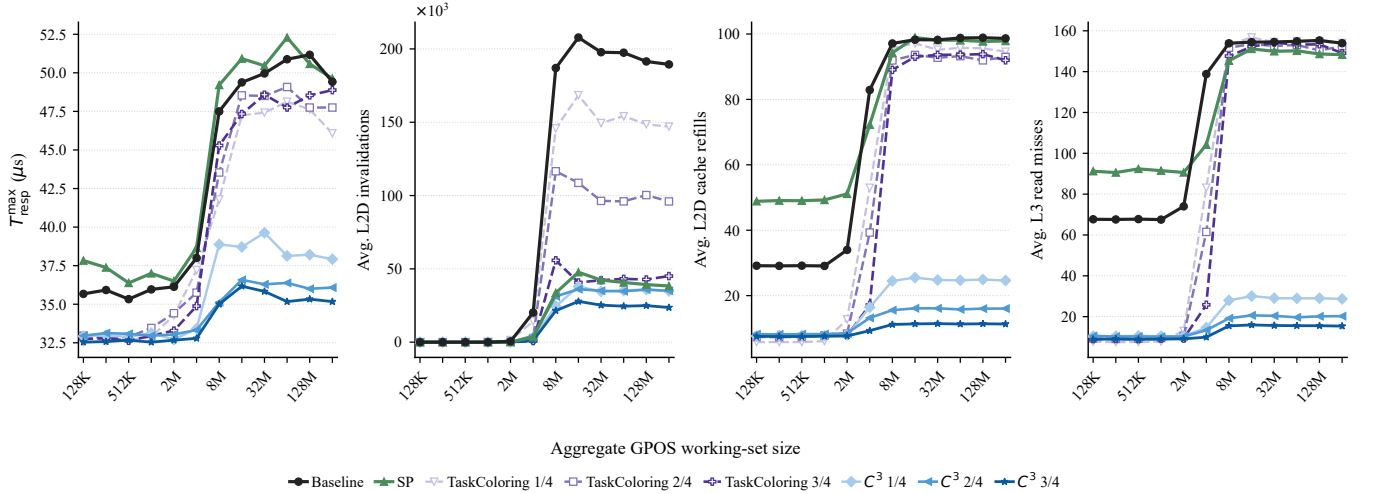


Fig. 5. RK3588 ablation and task-level allocation sensitivity at 10 Hz under combined interference. Fractions denote the proportion of B_T colors assigned to the critical region. Each point is obtained from one complete 600-s run.